

Phase 2 - Requirement Analysis

PocketSmart AI: Your Smart Budget & Recommendation Assistant

Date: 29 September 2026

1. Introduction

Requirement analysis defines what PocketSmart AI must provide to users and what the application must handle technically. Requirements are based on the three planners, authentication, recommendation engine, frontend pages and history.

2. Functional Goals

- Collect budget and contextual information.
- Generate category-specific recommendations.
- Support Home, Party and Jewelry planning.
- Support registration, login, logout and protected sessions.
- Support optional image input for jewelry.
- Display recommendations and recommendation history.

3. Users

The main users are people who want assistance planning purchases or events within a defined budget. The workflow is designed around selecting a category, entering relevant information and reviewing AI-generated suggestions.

Requirement Analysis - Functional Requirements

4. Home Interior Planner

- Enter total budget.
- Select room type such as Living Room, Kitchen or Bedroom.
- Enter quantities for required items.
- Process the request using Gemini.
- Return cost-effective recommendations for furniture, décor, lighting and related products.

5. Party Planner

- Enter total budget and guest count.
- Select event type and provide venue details.
- Allocate the budget across relevant categories.
- Suggest catering, decoration, entertainment, venue and accommodation options.

6. Jewelry Planner

- Enter budget, occasion and style preference.
- Optionally upload an outfit image.
- Use text and image context for matching.
- Return jewelry options aligned with occasion, outfit, style and budget.

Requirement Analysis - Authentication and Quality

7. Authentication Requirements

Route	Requirement
/register	Create new user account
/login	Authenticate user
/logout	Terminate session
/token	Issue JWT token
/session-info	Return session metadata
/session-data	Return session-specific information

8. Non-Functional Requirements

- Usability – simple and understandable forms.
- Security – protect credentials, API keys and session information.
- Reliability – validate input and handle insufficient AI results.
- Maintainability – separate routes, services, models, templates and configuration.
- Performance – process normal planner requests efficiently.

Requirement Analysis - Inputs, Outputs and Constraints

9. Main Inputs

Planner	Inputs
Home	Budget, room type, quantities, item requirements, preferences
Party	Budget, guest count, event type, venue details
Jewelry	Budget, occasion, style preference, optional outfit image
Authentication	Registration/login credentials

10. Main Outputs

- Structured AI-generated recommendations.
- Budget-aware category suggestions.
- Personalized jewelry suggestions when image/text context is supplied.
- Session-specific recommendation information.
- Recommendation history.

11. Constraints

- Gemini availability and response quality affect AI output.
- Product/service information may depend on APIs or simulated sourcing.
- Internet connectivity is required for cloud AI communication.
- Image recommendations depend on valid supported input.